

An Analysis of NFL Home Attendance Patterns, 2000 to 2019

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Introduction

This paper looks at game by game attendance for all 32 National Football League teams from 2000 through 2019. The question I set out to answer was simple. How does attendance behave across teams and over time, and are there patterns in it that a league or a single franchise would actually care about? Attendance matters because it is one of the clearest signs of fan engagement and a major source of revenue through tickets, concessions, and local sponsorship. If we can understand how it moves, we can understand something real about the business behind the game.

I focused on three questions. First, has league wide attendance grown, stayed flat, or fallen over these twenty years? Second, how steady is attendance from week to week inside a single season? Third, how does a team's attendance at home compare to the attendance it draws when it plays on the road? That last question turned out to be the most interesting, because a surprising number of teams draw bigger crowds away from home than they do in their own stadium. The rest of the paper walks through what I found and what I think is driving it.

Data

The data comes from a public NFL attendance dataset on Kaggle. The raw file has 10,846 team week records covering the 2000 through 2019 seasons for every team. Each record includes the team's city and name, the season, the week, that week's attendance, the season totals for home and away games, and the full season total. Weeks run from 1 through 17, which was the length of the NFL regular season during this period.

There are 638 missing weekly attendance values in the file. These are not random. They cluster in weeks 8 and 9 and line up with teams' bye weeks, when a team does not play. This is worth pointing out, because someone glancing at the gaps might read them as off season or as bad data, when really they are just scheduled byes. The dataset has no off season records at all. Every row is a game week during the season.

I did the analysis in SQL using MySQL, writing aggregation and self join queries to turn the raw records into team level and league level summaries. I then took those summary tables into Power BI and R to visualize them. The queries produced average weekly attendance by team and year, the league wide average by year, each team's home versus away attendance gap, and each week's share of the season total.

Methodology

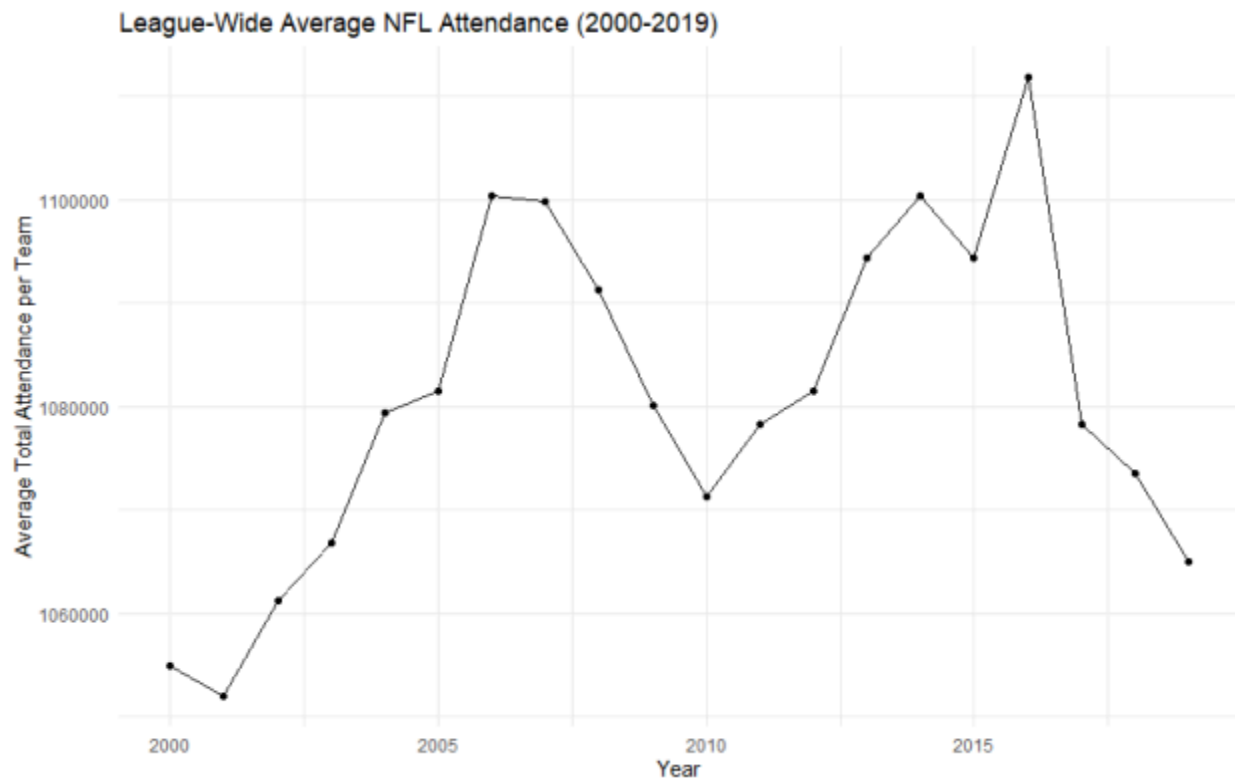
This is a descriptive analysis rather than a predictive one. Instead of fitting a regression model, my goal was to take a large panel of team week records and boil it down into summaries that are easy to read and easy to act on. I looked at the data along three lines. Time, meaning year over year. The season calendar, meaning week to week. And location, meaning home versus away.

The home versus away comparison is the heart of it. For each team I found the average home attendance and the average away attendance across all seasons, then took the difference as the gap. A positive gap means the team fills more seats in its own building than it draws on the road.

A negative gap means the opposite, that the team is actually a bigger draw as the visiting team than it is at home. This comparison is useful because home attendance is limited by a team's own stadium size, while away attendance reflects demand across a lot of different stadiums. The gap tells you something about both the local market and the national pull of a team.

Results

The first finding is that league wide attendance was basically flat over these twenty years, and then it fell. Average league attendance rose only about 1 percent between 2000 and 2019. It peaked in 2016 at roughly 1.11 million per team season and then dropped about 4 percent over the next three years. So the long run story is not growth. It is a mature market that had already leveled off and then started to soften at the end of the window. The chart below shows this trend.



The second finding is that inside a single season, attendance barely moves. Averaged across all teams and years, weekly attendance stayed in a tight band of roughly 66,000 to 69,000 from week 1 through week 17, a swing of only about 3 percent. There is no real early season or late season surge once you average it all out. NFL demand looks almost flat with respect to where you are in the season, which fits a league where most home games sell out no matter when they happen. I chose not to chart this one, because a flat line does not really show anything a sentence cannot say better.

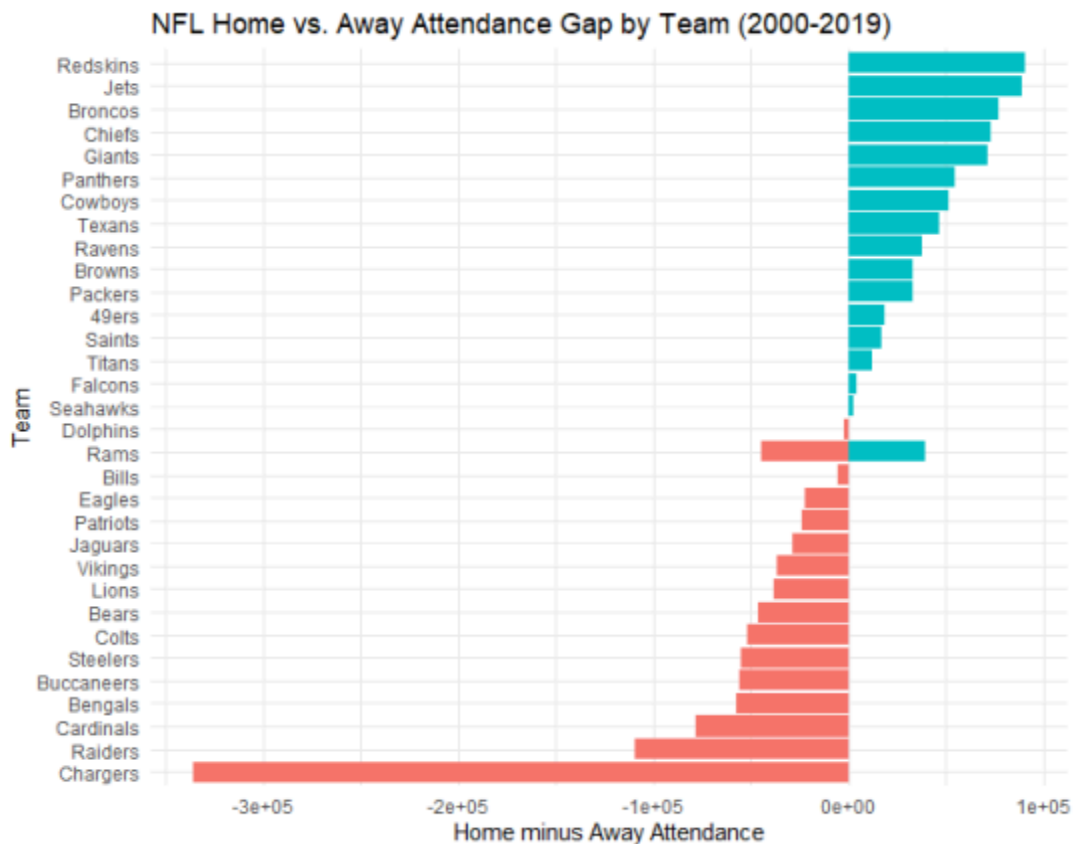
The third finding is the one I did not expect. Close to half of all teams draw bigger crowds on the road than they do at home. Out of the 34 team entries in the comparison, 17 had a negative gap.

The table below shows the teams at each extreme.

Team	Avg Home	Avg Away	Gap
Washington Redskins	639,074	548,084	+90,990
New York Jets	624,141	534,757	+89,385
Denver Broncos	607,991	531,050	+76,941
Kansas City Chiefs	598,235	525,065	+73,169
Arizona Cardinals	452,536	530,696	-78,161
Oakland Raiders	441,591	551,273	-109,681
LA Chargers*	239,613	542,909	-303,296

**The Los Angeles Chargers are a special case. After moving from San Diego, the team played in a temporary stadium of about 27,000 seats, which pushed their home number way down for reasons that have nothing to do with fan demand.*

The split is notable. Big market or big stadium teams like the Washington Redskins and the New York Jets fill far more of their own seats than they draw on the road. At the other end, teams like the Cardinals and the Raiders pull noticeably bigger crowds in other teams' stadiums than in their own. A chart makes the full spread easy to see across all the teams at once.



The reason so many teams draw better on the road probably comes down to stadium size. When a team plays away it's filling seats in whatever stadium it visits so it shows a high demand for that team across the nation. At home a team can only sell as many seats as its own building holds so the negative gap more likely shows a smaller older stadium than a weak fan base. The Chargers are the clearest example. From 2017 to 2019 they played home games at Dignity Health Sports Park, which only holds 27,000 seats, while their permanent stadium at SoFi was being built.

Limitations

There are a few things this analysis does not account for, and they would all need attention before anyone drew firm conclusions from it. The biggest one is stadium capacity. A negative home gap might just mean a team has a smaller building, since a sold out small stadium still records low attendance. Team performance is another factor, because winning seasons probably lift both home and away crowds, and I did not control for wins here. Market size and relocation matter too, since the Chargers and the Rams both moved to Los Angeles during this window and that distorts their numbers for those years.

Another limitation is that these are reported attendance figures, which, for the NFL, usually mean tickets distributed rather than a count of people who actually came to the gate. That means a game could be listed as a sellout but still might have empty seats. This could be due to season ticket holders not going to every game so the numbers may be a little high compared to actual game attendance.

Conclusion

Across twenty seasons, NFL attendance tells a pretty consistent story of a saturated market. League wide numbers were flat and then softened after 2016, and demand inside a season hardly moved from week to week. The finding I would actually build on is the home versus away gap, because it shows that how well a team fills its own stadium is a different question from how much of a draw that team is nationally, and the two often point in opposite directions. For a league or a team thinking about stadium investment, ticket pricing, or where to put a franchise, separating local capacity from national appeal is a more useful way to look at things than headline attendance on its own.